

GRAIN DIRECTION

SAME SIZE ARTWORK
LEAFLET: 225 mm x 270 mm
FOLDING SIZE: 38 mm x 135 mm

38 mm		38 mm		38 mm		38 mm		38 mm	
SPACE FOR PHARMACODE		SPACE FOR PHARMACODE		SPACE FOR PHARMACODE		SPACE FOR PHARMACODE		SPACE FOR PHARMACODE	
135 mm									
25 mm									
<i>For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory</i> Combiwave FB200 Formoterol Fumarate 6mcg/actuation and Budesonide 200mcg/actuation Pressurised Inhalation COMPOSITION Each actuation delivers: Formoterol Fumarate Dihydrate BP equivalent to Anhydrous Formoterol Fumarate..... 6 mcg Budesonide BP..... 200 mcg Excipients & Propellant 1,1,1,2 Tetrafluoro Ethane (HFA 134a).....g.s. DOSAGE FORM Inhalation aerosol DESCRIPTION Formoterol Fumarate and Budesonide Pressurised Inhalation is a combination of budesonide. White to off white color suspension filled in aluminium aerosol can with silver ferrule valve provided with actuator suitable for oral inhalation when actuated a metered spray is produced. PHARMACOLOGY Pharmacodynamics Formoterol Fumarate and Budesonide Pressurised Inhalation contain both budesonide and formoterol; therefore, the mechanisms of action described below for the individual components apply to Formoterol Fumarate and Budesonide Pressurised Inhalation. These drugs represent two classes of medications (a synthetic corticosteroid and a long-acting, selective beta2- adrenoceptor agonist) that have different effects on the clinical, physiological, and inflammatory indices of asthma. Budesonide: Budesonide is an anti-inflammatory corticosteroid that exhibits potent glucocorticoid activity and weak mineralocorticoid activity. Inflammation is an important component in the pathogenesis of asthma. Corticosteroids have a wide range of inhibitory activities against multiple cell types (e.g., mast cells, eosinophils, neutrophils, macrophages, and lymphocytes) and mediators (e.g., histamine, eicosanoids, leukotrienes, and cytokines) involved in allergic and non-allergic-mediated inflammation. These anti-inflammatory actions of corticosteroids may contribute to their efficacy in asthma. Studies in asthmatic patients have shown a favorable ratio between topical anti-inflammatory activity and systemic corticosteroid effects over a wide range of doses of budesonide. This is explained by a combination of a relatively high local anti- inflammatory effect, extensive first pass hepatic degradation of orally absorbed drug (85%-95%), and the low potency of formed metabolites. Formoterol Fumarate: Formoterol fumarate is a long-acting, selective beta2- adrenergic agonist with a rapid onset of action. Inhaled formoterol fumarate acts locally in the lungs as a bronchodilator. In vitro studies have shown that formoterol has over 200-fold greater agonist activity at beta2- receptors than at beta1-receptors. The in vitro binding selectivity to beta2-adrenoceptors over beta1- adrenoceptors is higher for formoterol than for salbutamol (5 times), whereas salmeterol has a higher (3 times) beta2-selectivity ratio than formoterol. Although beta2-receptors are the predominant adrenergic receptors in bronchial smooth muscle and beta1- receptors are the predominant receptors in the heart, there are also beta2-receptors in the human heart, which comprise 10-50% of the total beta-adrenergic receptors. The precise function of these receptors has not been established, but they raise the possibility that even highly selective beta2-agonists may have cardiac effects. The pharmacological effects of beta2-adrenoceptor agonist drugs, including formoterol, are, at least in part, attributable to the stimulation of intracellular adenylyl cyclase, the enzyme that catalyses the conversion of adenosine triphosphate (ATP) to cyclic-3', 5'-adenosine monophosphate (cyclic AMP). Increased cyclic AMP levels cause relaxation of bronchial smooth muscle and inhibit the release of mediators of immediate hypersensitivity from the cells, especially from mast cells. Pharmacokinetics Absorption: Budesonide: Orally inhaled budesonide is rapidly absorbed in the lungs. Formoterol: Rapidly absorbed after inhalation. Distribution: Budesonide: 85-90% bound to plasma proteins. Metabolism: Formoterol: Primary metabolism pathway is via direct glucuronidation and O-demethylation followed by conjugation to inactive metabolites. Excretion: Budesonide: Excreted in urine and faeces as metabolites. Formoterol: Excreted mainly in urine and some in faeces. Special Populations: Geriatric: The pharmacokinetics of Formoterol Fumarate and Budesonide Pressurised Inhalation in geriatric patients has not been specifically studied. Paediatric: In age 6-11 years, peak budesonide concentrations of 1.4 nmol/L occurred at 20 minutes post-dose. Approximately 3.5% of the delivered formoterol dose was recovered in the urine as unchanged formoterol. Renal or Hepatic Insufficiency: There are no data regarding the specific use of Formoterol Fumarate and Budesonide Pressurised Inhalation in patients with hepatic or renal impairment. Reduced liver function may affect the elimination of corticosteroids. Budesonide pharmacokinetics was affected by compromised liver function as evidenced by a doubled systemic availability after oral ingestion. The intravenous budesonide pharmacokinetics was, however, similar in cirrhotic patients and in healthy subjects. Specific data with formoterol is not available, but because formoterol is primarily eliminated via hepatic metabolism, an increased exposure can be expected in patients with severe liver impairment. INDICATIONS Regular treatment of persistent asthma where use of combination of inhaled corticosteroid therapy plus a bronchodilator has been found to be appropriate. In patients not adequately controlled with inhaled corticosteroid and 'as needed' inhaled short acting beta2- agonist or patients already adequately controlled on both inhaled corticosteroids and long acting beta2-agonist. DOSAGE AND ADMINISTRATION For oral inhalation only. Asthma Treatment of asthma in patient's ≥12 years: 2 inhalations twice daily of Formoterol Fumarate and Budesonide Pressurised Inhalation. Starting dosage is based on asthma severity. COPD Maintenance treatment of airflow obstruction in COPD: 2 inhalations of Formoterol Fumarate and Budesonide Pressurised Inhalation twice daily. CONTRAINDICATIONS Formoterol Fumarate and Budesonide Pressurised Inhalation is contraindicated in patients with a history of hypersensitivity to any of the components of the drug product. This drug product is contraindicated in the primary treatment of status asthmaticus or other acute episodes of asthma or COPD where intensive measures are required. WARNINGS AND PRECAUTIONS Asthma-related death: Long-acting beta2-adrenergic agonists increase the risk. Prescribe only for recommended patient populations. Deterioration of disease and acute episodes: Do not initiate in acutely deteriorating asthma or to treat acute symptoms. Use with additional long-acting beta2-agonist: Do not use in combination because of risk of overdose. Localized infections: Candida albicans infection of the mouth and throat may occur. Monitor patients periodically for signs of adverse effects on the oral cavity. Advise patients to rinse the mouth following inhalation. Pneumonia: Increased risk in patients with COPD. Monitor patients for signs and symptoms of pneumonia and other potential lung infections. Immunosuppression: Potential worsening of infections (e.g., existing tuberculosis, fungal, bacterial, viral, or parasitic infection; or ocular herpes simplex). Use with caution in patients with these infections. More serious or even fatal course of chickenpox or measles can occur in susceptible patients. Transferring patients from systemic corticosteroids: Risk of impaired adrenal function when transferring from oral steroids. Taper patients slowly from systemic corticosteroids if transferring to Formoterol Fumarate and Budesonide Pressurised Inhalation. Hypercorticism and adrenal suppression: May occur with very high dosages or at the regular dosage in susceptible individuals. If such changes occur, discontinue Formoterol Fumarate and Budesonide Pressurised Inhalation slowly. Strong cytochrome P450 3A4 inhibitors (e.g., ritonavir): Risk of increased systemic corticosteroid effects. Exercise caution when used with Formoterol Fumarate and Budesonide Pressurised Inhalation. Paradoxical bronchospasm: Discontinue Formoterol Fumarate and Budesonide Pressurised Inhalation and initiate alternative therapy if paradoxical bronchospasm occurs. Patients with cardiovascular or central nervous system disorders: Use with caution because of beta- adrenergic stimulation. Decreases in bone mineral density: Assess bone mineral density initially and periodically thereafter. Effects on growth: Monitor growth of pediatric patients. Glaucoma and cataracts: Close monitoring is warranted. Metabolic effects: Be alert to eosinophilic conditions, hypokalemia and hyperglycemia. Coexisting conditions: Use with caution in patients with convulsive disorders, thyrotoxicosis, diabetes mellitus, and ketoacidosis. Drug Interactions Concurrent administration of Formoterol Fumarate and Budesonide Pressurised Inhalation and other drugs, such as short-acting beta2-agonists, intranasal corticosteroids, and antihistamines/decongestants has not resulted in an increased frequency of adverse reactions. No formal drug interaction studies have been performed with Formoterol Fumarate and Budesonide Pressurised Inhalation. Ketoconazole, a potent inhibitor of cytochrome P450 (CYP) isoenzyme 3A4 (CYP3A4), the main metabolic enzyme for corticosteroids, increased plasma levels of orally ingested budesonide. At recommended doses, cimetidine had a slight but clinically insignificant effect on the pharmacokinetics of oral budesonide. Specific drug- drug interaction studies with formoterol have not been performed. Pregnancy There are no adequate and well-controlled studies of Formoterol Fumarate and Budesonide Pressurised Inhalation in pregnant women. Formoterol Fumarate and Budesonide Pressurised Inhalation was teratogenic and embryocidal in rats. Budesonide alone was teratogenic and embryocidal in rats and rabbits, but not in humans at therapeutic doses. Formoterol fumarate alone was teratogenic in rats and rabbits. Formoterol fumarate was also embryocidal, increased pup loss at birth and during lactation, and decreased pup weight in rats. Formoterol Fumarate and Budesonide Pressurised Inhalation should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus. PE00000 MY ICONGRAPHICS CODE: PANTONE SHADE PANTONE BLACK PROCESS C Supersedes Artwork Code PE23880 PHARMACODE:									

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Lactation

Since there are no data from controlled trials on the use of Formoterol Fumarate and Budesonide Pressurised Inhalation by nursing mothers, a decision should be made whether to discontinue nursing or to discontinue Formoterol Fumarate and Budesonide Pressurised Inhalation, taking into account the importance of Formoterol Fumarate and Budesonide Pressurised Inhalation to the mother.

UNDESIRABLE EFFECTS

As Combiwave FB contains formoterol and budesonide, the type and severity of adverse reactions associated with each of the compounds may be expected. There is no incidence of additional adverse events following concurrent administration of the two compounds. As with other inhalation therapy, paradoxical bronchospasm may occur.

Adverse events associated with Formoterol: Tremor, palpitations and headache have been reported, but tend to be transient and reduce with regular therapy. Cardiac arrhythmias, muscle cramps and hypokalaemia may occur in some patients.

Budesonide: Hoarseness and candidiasis (thrush) of the mouth and throat can occur in some patients; which may be relieved by gargling with water after use of Combiwave FB. Symptomatic candidiasis can be treated with topical anti-fungal therapy whilst still continuing with Combiwave FB.

OVERDOSAGE

An overdose of formoterol would likely lead to effects that are typical for beta₂-adrenergic agonists: tremor, headache, palpitations. Symptoms reported from isolated cases are tachycardia, hyperglycaemia, hypokalaemia, prolonged QTinterval, arrhythmia, nausea and vomiting. Supportive and symptomatic treatment may be indicated. A dose of 90 micrograms administered during three hours in patients with acute bronchial obstruction raised no safety concerns.

Acute overdosage with budesonide, even in excessive doses, is not expected to be a clinical problem. When used chronically in excessive doses, systemic glucocorticosteroid effects, such as hypercorticism and adrenal suppression, may appear.

If Formoterol Fumarate and Budesonide Pressurised Inhalation therapy has to be withdrawn due to overdose of the formoterol component of the drug, provision of appropriate inhaled corticosteroid therapy must be considered.

STORAGE:

Store below 30°C. Do not freeze.
Keep all medicines out of reach of children.

PACKAGING INFORMATION

Formoterol Fumarate and Budesonide Pressurised Inhalation (6 mcg + 200 mcg per actuation)
Each sales pack is available as a canister containing 150 metered doses.

SHELF-LIFE

24 months

MAL No.: MAL14095024AZ

INSTRUCTIONS FOR PATIENT FOR USING THE INHALER WITHOUT SPACER

Important: Follow instructions carefully.

Instructions for Use Important: Testing your inhaler:

If the inhaler is new or has not been used for seven days or more, one puff should be released into the air (See pic no. 1). It is not necessary to shake the inhaler before use because this is a solution aerosol.



1

Remove the cap from the mouthpiece; the mouthpiece should be inspected for the presence of foreign objects before each use. Make sure the canister is fully and firmly inserted into the actuator. (See pic no. 2)



2

Hold the inhaler upright with your thumb on the base. Place out fully through your mouth expelling as much air from your lungs as possible. Thereafter, place the mouthpiece of the inhaler in your mouth between your teeth. (See pic no. 3)



3

Close your lips around it (do not bite it) tilt your head slightly backwards. Start breathing in slowly through your mouth. As you breathe in steadily and deeply, press down the canister to release one puff. (See pic no. 4)



4

Remove the inhaler from your mouth and hold breath for 5-10 seconds or for as long as it is comfortable. Breathe out slowly. (See pic no. 5)



5

Note:

If another puff is required wait for at least one minute and repeat steps 2 through 5 for each puff prescribed by your Physician. Rinse your mouth or gargle with water after inhaling your recommended dose (Single dose means total number of puffs advised by the Physician at once). This is likely to reduce the soreness that may be caused by the drug. After use, replace the mouthpiece cover. (See pic no. 6)



6

Practice in front of the mirror for the first few times. If you see 'mist' coming from top of the inhaler or sides of the mouth, this indicates failure of technique. Start again from Step 2. (See pic no. 7)



7

For Children:
Children should use the inhaler under adult supervision, as instructed by the Physician. (See pic no. 8)



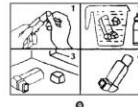
8

Cleansing:

Clean the inhaler at once in a week to prevent any blockage and to carefully follow the instructions on cleaning below.

1. Gently pull the metal canister out of the plastic body of the inhaler. Remove the mouthpiece cover.
2. Rinse the plastic body and the mouthpiece cover in warm / running water.
3. Dry thoroughly. Avoid excess heat.
4. Replace the canister and the mouthpiece cover correctly.

Discard the inhaler along with the canister after using the labelled number of inhalation.



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Date of Revision: 1 October 2018

PRODUCT REGISTRATION HOLDER

Glenmark Pharmaceuticals (Malaysia) Sdn Bhd (660397-W)
D-31-02, Menara Suezcap 1, No. 2, Jalan Kerinchi,
59200 Kuala Lumpur, Malaysia

Manufactured by:

glenmark PHARMACEUTICALS LTD.
(Unit III), Village Kishanpura,
Baddi-Nalagarh Road, Tehsil Baddi,
Distt. Solan (H.P.) 173 205.

PE00000 MY

ICONGRAPHICS CODE

PANTONE SHADE



PANTONE BLACK PROCESS C

Supersedes
Artwork Code
PE23880

PHARMACODE: